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Assignment - 3

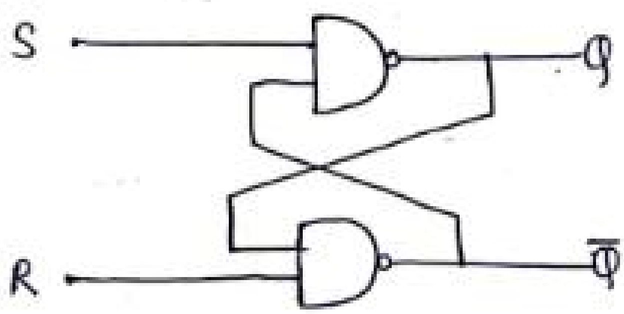
Ques 1. Explain the difference between a latch and a flip-flop.

Sol^n → **Latch** :- latch is a digital circuit which converts its output according to its inputs instantly.
latches can store a single bit of information and ~~latches~~ hold its value until it is updated by new input signals.

→ Asynchronous :- does not depend on clock pulse.

→ Uncontrolled.

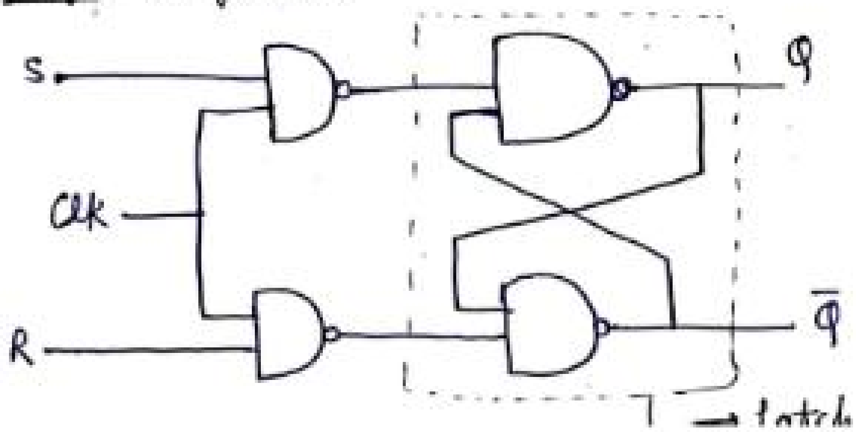
Logic Diagram :-



Flip Flop :- The memory elements used in clocked sequential circuits are called Flip-Flop.
These circuits store one bit of information. A flip-flop circuit has two outputs, one for the normal value and one for the complement value of the bit stored in it.

→ A flip-flop circuit can maintain a binary state indefinitely until directed by an input signal to switch states.

logic diagram :-

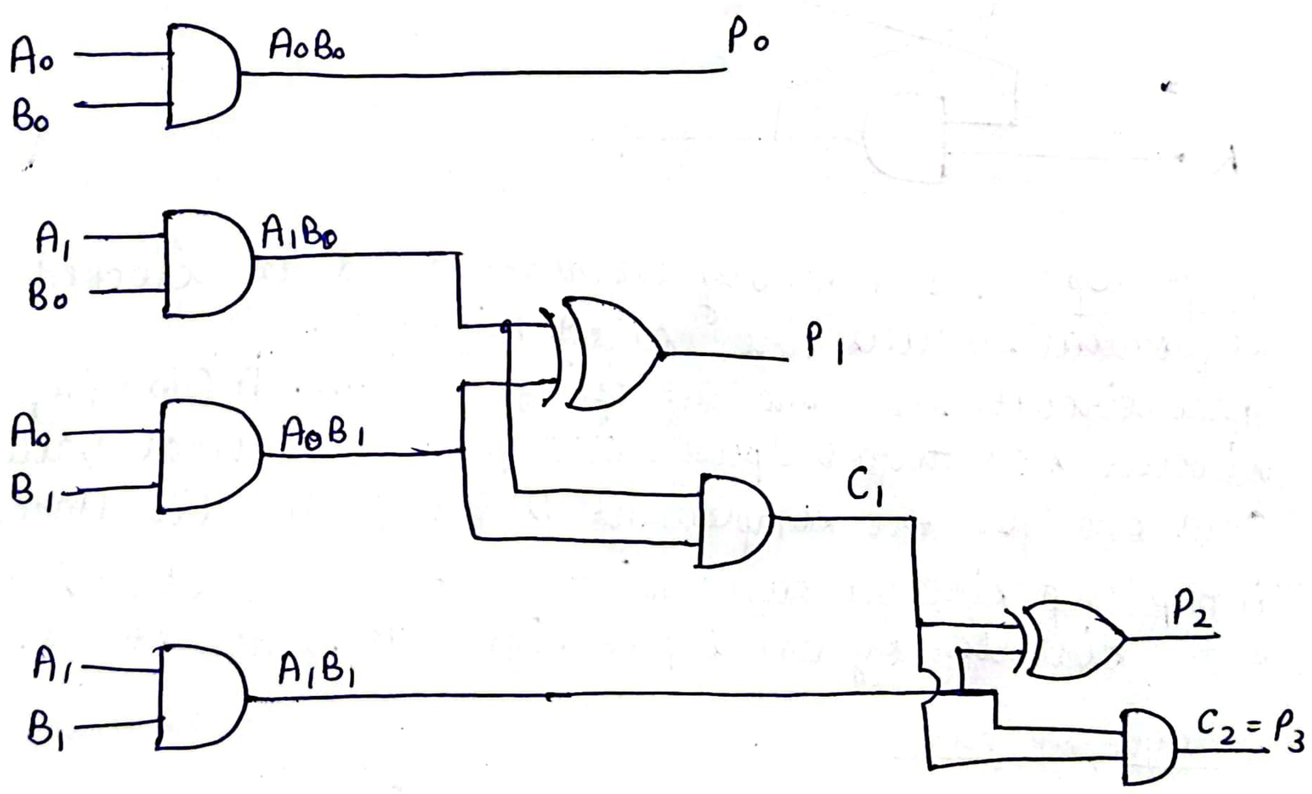


Ques 2⇒ Design a circuit of 2-bit binary multiplier.

Sol^n: let $A = A_1 A_0$
$B = B_1 B_0$

		A_1	A_0
		$\times$ B_1	B_0
	C_1	A_1B_0	A_0B_0
	A_1B_1	A_0B_1	X
C_2	$A_1B_1 + C_1$	$A_1B_0 + A_0B_1$	A_0B_0

$P_0 = A_0B_0$
$P_1 = A_1B_0 + A_0B_1$
$P_2 = A_1B_1 + C_1$
$P_3 = C_2$

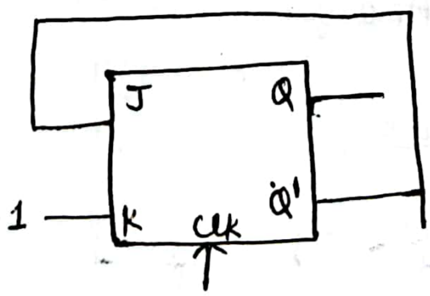


Ques 4:- What are the differences between asynchronous (ripple) counters and synchronous counters?

Soln: Counters are of two types depending upon clock pulse applied. These ~~containers~~ counters are Asynchronous counter and Synchronous counter. In Asynchronous counter is also known as Ripple counter, different flip-flops are triggered with different clock, not simultaneously. While in Synchronous counter, all flip flops are triggered with same clock simultaneously and Synchronous Counter is faster than asynchronous counter in operation. Let's see the difference between these two counters:-

Synchronous Counter	Asynchronous Counter
→ In synchronous counter, all flip flops are triggered with same clock simultaneously.	→ In asynchronous counter, different flip flops are triggered with different clock, not simultaneously
→ Synchronous counter is faster than asynchronous counter in operation.	→ Asynchronous Counter is slower than Synchronous Counter in operation
→ Synchronous counter is also called Parallel Counter.	→ Asynchronous counter is also called Serial Counter.
→ Synchronous Counter designing as well as implementation are complex due to increasing the number of states.	→ Asynchronous Counter designing as well as implementation is very easy
→ Synchronous Counter will operate in any desired count sequence.	→ Asynchronous Counter will operate only in fixed count sequence.

Ques 3. In a J-K flip flop, we have J=Q' and K=1. Assuming the flip-flop was initially cleared and then for 6 pulses. Analyse the sequence at Q output will be.



Ans: Characteristic Table of JK Flip Flop is:-

J	K	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	$\bar{Q}_n$

Analysis;

$J = \bar{Q}, \; K = 1$

where $Q = 0, \; \bar{Q} = 1 \Rightarrow J = 1$

$\boxed{Q_{n+1} = \bar{Q}_n = 1}$

when $Q = 1, \; \bar{Q} = 0 \Rightarrow J = 0$

$\boxed{Q_{n+1} = 0}$

So, the sequence at Q output will be 0 1 0 1 0 1

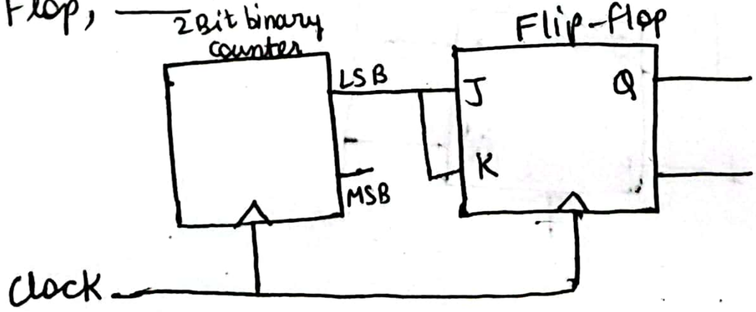
Ques 5: Define state reduction in context of clocked sequential circuits and explain its significance.

Ans: State reduction is a technique used to minimize the number of states in a sequential circuit while preserving its functionality. This optimization is ~~ctr~~ crucial for designing efficient and cost effective digital systems.

Importance of State Reduction:-

- Reduced Hardware Complexity :- Fewer states directly translate to fewer flip-flop which are fundamental building blocks of sequential circuits. This reduction can lead to ~~similar~~ smaller cheaper circuits.
- Improved performance :- A smaller number of state can simplify the logic required to implement the state transition and output functions.

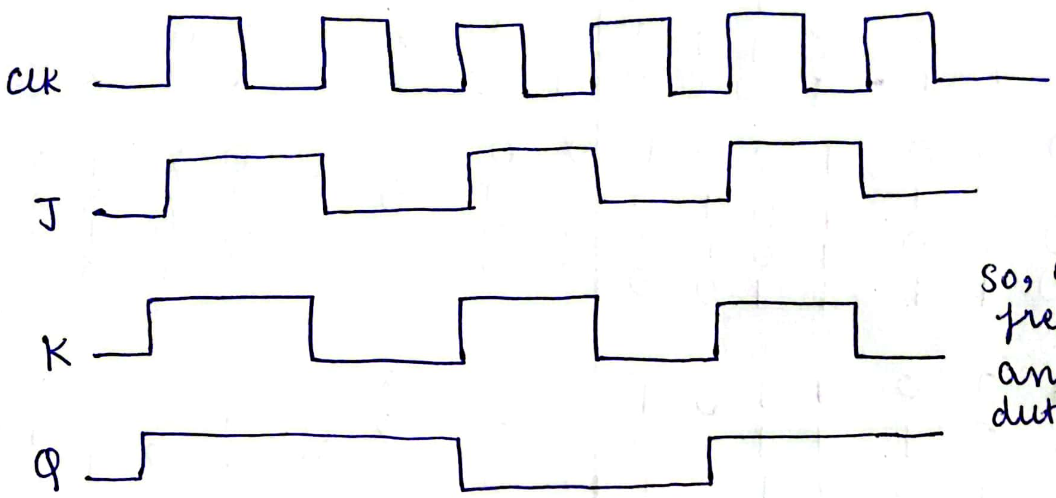
Ques 6: For the circuit shown, the clock frequency is fo and the duty cycle is 25%. Analyse the signal at the Q output of the Flip-Flop,



Ans 6. Frequency of clock is fo
Input J = K = LSB say y
Output of JK Flip flop →

J	K	Q_{n+1}
0	0	Q_n
1	1	$\overline{Q_n}$

For JK flip-flop, 00 will not change the state.
So, output frequency = $\frac{f_o}{2}$, duty cycle = 50%.



So, output frequency = fo/2 and duty cycle = 50.

Ques 7→ Design a 4-bit binary synchronous counter using JK flip flops. Prove the state table and logic diagram.

Ans JK Flip flop excitation Table :-

Q	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

For 4-bit Binary synchronous counter 4 JK flip-flops are required. Let $J_0 K_0$, $J_1 K_1$, $J_2 K_2$, $J_3 K_3$ be the 8 inputs of the 4 J-K flip flops.

Present state				Next state				F_3		F_2		F_1		F_0	
Q_3	Q_2	Q_1	Q_0	Q_4	Q_2	Q_1	Q_0	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	0	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	1	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	X	0	1	X
0	1	1	1	1	0	0	0	1	X	X	1	X	1	X	1
1	0	0	0	1	0	0	1	X	0	0	X	0	X	1	X
1	0	0	1	1	0	1	0	X	0	0	X	1	X	X	1
1	0	1	0	1	0	1	1	X	0	0	X	X	0	1	X
1	0	1	1	1	1	0	0	X	0	1	X	X	1	X	1
1	1	0	0	1	1	0	1	X	0	X	0	0	X	1	X
1	1	0	1	1	1	1	0	X	0	X	0	1	X	X	1
1	1	1	0	1	1	1	1	X	0	X	0	X	0	1	X
1	1	1	1	0	0	0	0	X	1	X	1	X	1	X	1

Expression :-

$J_0 = 1$, $K_0 = 1$

for J_1

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00		1	X	X
01		1	X	X
11		1	X	X
10		1	X	X

$\boxed{J_1 = Q_0}$

for K_1

$Q_3Q_2 \backslash Q_1Q_0$				
	X	X	1	
	X	X	1	
	X	X	1	
	X	X	1	

$\boxed{K_1 = Q_0}$

for J_2

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00			1	
01	X	X	X	X
11	X	X	X	X
10			1	

$J_2 = Q_1Q_0$

for K_2

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	X	X
01			1	
11			1	
10	X	X	X	X

$K_2 = Q_1Q_0$

for J_3

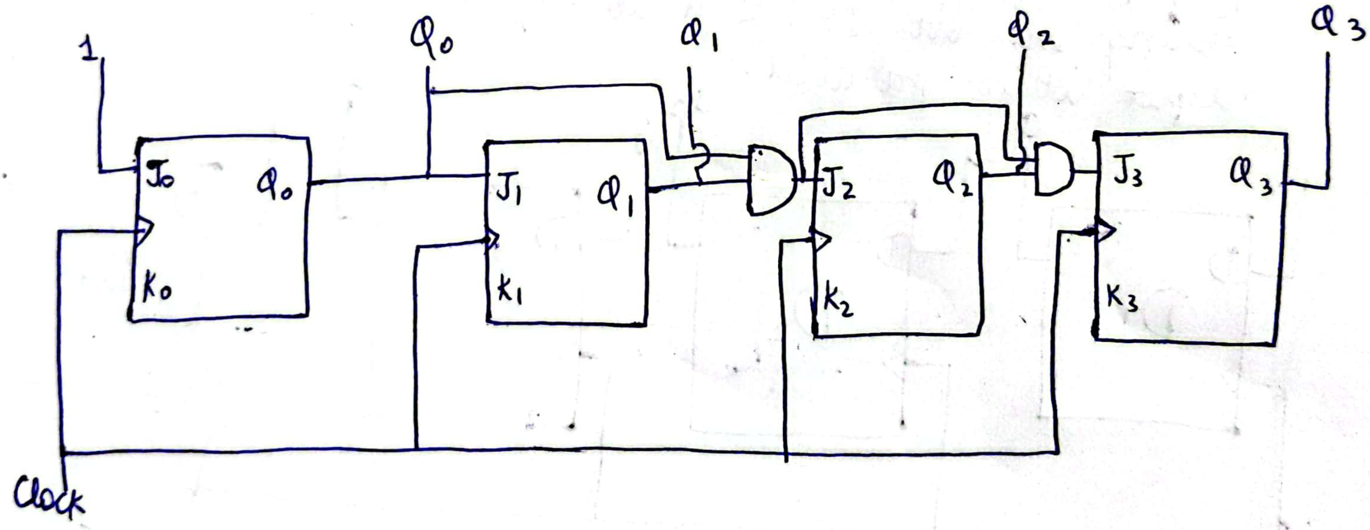
$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00				
01			1	
11	X	X	X	X
10	X	X	X	X

$J_3 = Q_2Q_1Q_0$

for K_3

$Q_3Q_2 \backslash Q_1Q_0$	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11			1	
10				

$K_3 = Q_2Q_1Q_0$



Ques 8. Explain how don't care conditions are used in the state assignment for sequential circuits. Provide an example.

Ans 8. Don't care conditions are used in state assignment for sequential circuits to simplify input functions to flip flop or other circuits. This is possible when a sequential circuits uses fewer states than maximum number of states availables

Example :- Let consider a simple sequential with 5 states A, B, C, D, E. We need to assign binary values to these.

A → 000
B → 001
C → 010
D → 011
E → 100

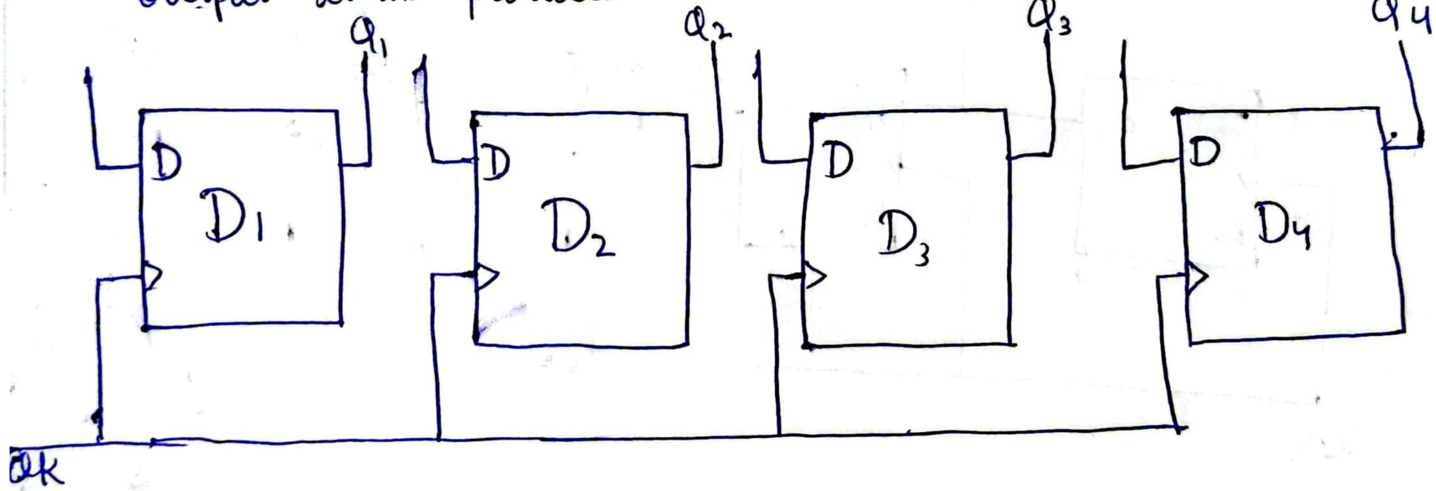
In this case binary codes 101 and 110, and 111 are unused. These states can be treated as don't cares during design process.

Next State = $A'B'C' + A'BC + AB'C + ABC' + ABC$

K-map → Next State = $A'B' + AB = (\overline{A \oplus B})$

Ques 9. Design a 4-bit parallel-in-parallel out shift register using D-flip flop. Include the circuit diagram and explain the operation.

Ans 9. A parallel-in-parallel out shift register is a collection of flip-flop arranged in series, with each capable of storing one bit. It has a ability to load data and output it in parallel.

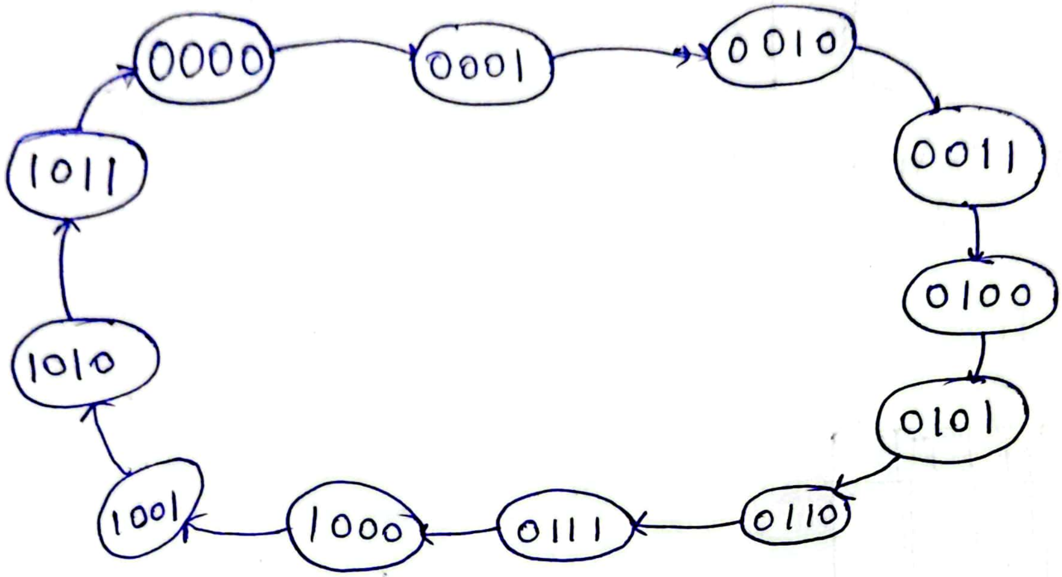


A 4-bit shift register consist of four flip flop D_0, D_1, D_2, D_3. Each flip flop stores a bit. The data can be loaded in flip-flop simultaneous through the parallel input, known as data input. One data is loaded it can be read out simultaneously from each flip flop ~~to shift~~ through parallel outputs.
Each flip flop contents are passed sequentially to the following flip-flop to shift data out. This can be accomplished by clock signal that triggers shifting opearation.

Ques 10, Design and implement a module-12 synchronous counter using T flip flop. Show all steps, state diagram and logic circuit.

Ans 10, For designing a modulo-12 synchronous counter atleast 4 flip flops are required

State diagram :-



State Table :-

Present State Q_3	Q_2	Q_1	Q_0	Next State Q_3	Q_2	Q_1	Q_0	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	1	0	1	0	0	0	1	1
1	0	1	0	1	0	1	1	0	0	0	1
1	0	1	1	0	0	0	0	1	0	1	1

Expressions:-

$T_0 = 1$

For T_3

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00				
01			1	
11	x	x	x	x
10			1	

$T_3 = Q_2Q_1Q_0 + Q_3Q_1Q_0$
$= Q_1Q_0(Q_3 + Q_2)$

For T_2

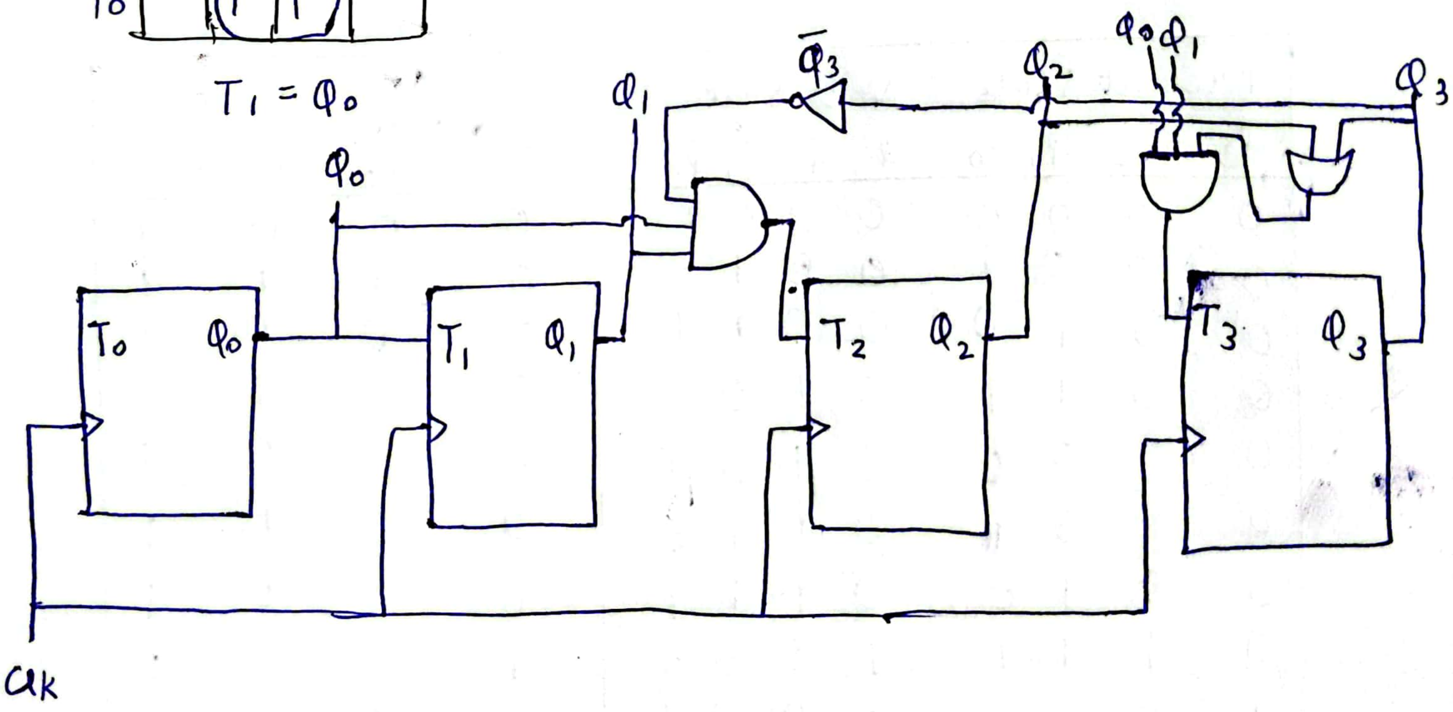
Q_3Q_2 \ Q_1Q_0	00	01	11	10
00			1	
01			1	
11	x	x	x	x
10				

$T_2 = \overline{Q}_3 Q_1 Q_0$

For T_1

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00		1	1	
01		1	1	
11	x	x	x	x
10		1	1	

$T_1 = Q_0$



Ques 11 :- Reduce the number of state table and tabelate the state table.

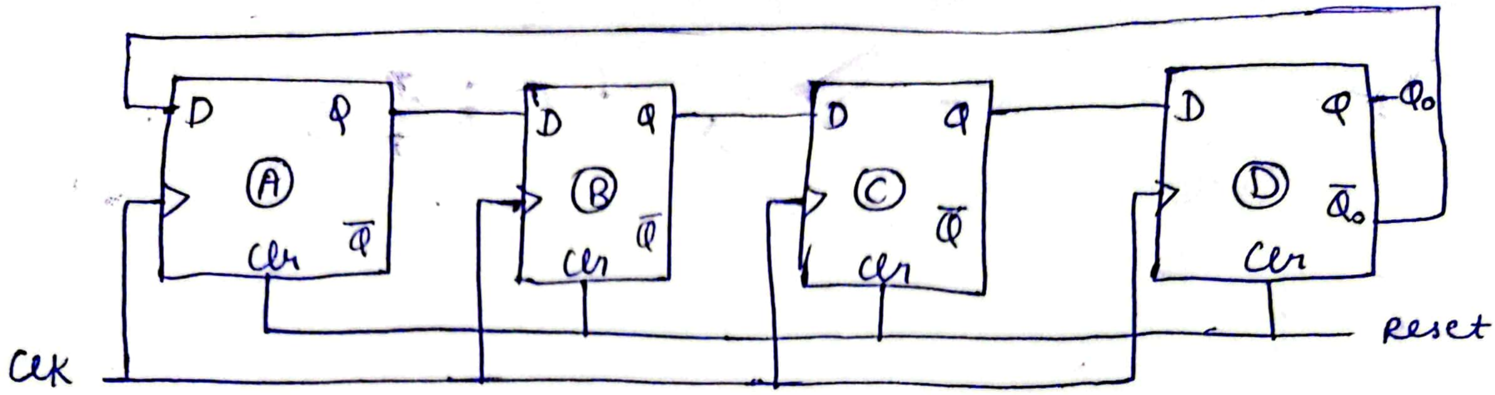
Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Reduced State Table :-

Present State	Next State $x=0$	Next State $x=1$	Output $x=0$	Output $x=1$
a	f	b	0	0
b	d	a	0	0
d	g	a	1	0
f	f	b	1	1
g	g	d	0	1

Ques 12 → What is Johnson counter? Design a 4-bit Johnson counter and explain in state sequence.

Ans, A Johnson counter also known as twisted ring counter is a type of synchronous sequential logic circuit that produces a unique state sequence. It is a modification of ring counter where the inverted output of the last flip flop is feel track to input of first flip flop.



State Sequence :-

1. Initial State :- All flip flop reset to (0000)
2. State 1 :- The first flip flop toggles to 1 (1000)
3. State 2 :- The second flip flop toggles to 1 (1100)
4. State 3 :- The third flip flop toggles to 1 (1110)
5. State 4 :- The all flip flop sets to 1. (1111)
6. State 5 :- The fourth flip flop toggles to 0 (1110)
7. State 6 :- The third flip flop toggles to 0 (1100)
8. State 7 :- The second flip flop toggles to 0 (1000)
9. State 8 :- The first flip flop toggles to 0, returning initial state (0000)